Scenario 1 (Full Marks - Neat & Legible Handwriting)

Ans 1.

The OSI model has 7 layers:

1. Physical Layer: Transmits raw bits over cables/media.
2. Data Link Layer: Handles node-to-node transfer & MAC framing.
3. Network Layer: Manages IP routing and packet forwarding.
4. Transport Layer: End-to-end delivery & error checking (e.g. TCP).
5. Session Layer: Manages sessions between applications.
6. Presentation Layer: Syntax, encryption & data formatting.
7. Application Layer: User interface & protocols like HTTP.

Scenario 2 (Partial Credit - Incomplete Steps & Stops Mid-way)

Ans 2.

Given matrix $A = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$

Step 1: Find Determinant

$$\det(A) = (4 \times 6) - (7 \times 2) = 24 - 14 = 10$$

Since $\det(A) = 10 \neq 0$, inverse exists.

Step 2: Adjoint of matrix

$$\text{adj}(A) = \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

(Leave the page blank here without calculating the final inverse)

Scenario 3 (Skipped / Blank Answer)

Ans 3.

Scenario 4 (Messy / Rushed Cursive - Correct Content)

Ans 4.

Binary Search:

Precondition: array must be sorted.

Working: low = 0, high = n - 1.

mid = (low + high) / 2

if arr[mid] == key → return mid

if key < arr[mid] → high = mid - 1

else → low = mid + 1

Repeat while low <= high.

Scenario 5 (Completely Illegible / Scribble)

Ans 5.

Light rxn takes plce in thylkd mmbrne ...

chlrofl absobs phtns ...

wtr splts $2H_2O \rightarrow$ ...

elcctr trnsprt chn mkng ATP & NDPH ...

rleses Oxgn.